

1. Refer to Section 10.3 in CLRS4 that gives a scheme for representing trees in which nodes may have more than 2 children. This scheme is referred to as *left-child, right-sibling representation*. Give an $\mathcal{O}(n)$ -time procedure that prints out all the keys of an arbitrary rooted tree with n nodes, where the tree is stored using the left-child, right-sibling representation.
2. The left-child, right-sibling representation of an arbitrary rooted tree uses three pointers in each node: left-child, right-sibling, and parent. From any node, its parent can be accessed in constant time and all its children can be accessed in time linear in the number of children. Show how to use only two pointers and one boolean value in each node so that x 's parent or all of x 's children can be accessed in time linear in the number of x 's children.
3. Give an algorithm that interprets the computation specified by a given arithmetic expression and performs the computation to arrive at the result without using expression trees. Here, the arithmetic expressions are assumed to be fully parenthesized.
4. Describe an algorithm to convert an expression in postfix form into an expression tree.
5. Describe an algorithm that substitutes all nodes of an expression tree with a specific value with a different value.